

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech. (CE) Examination****March-April 2018****CE412: EMBEDDED SYSTEM & VLSI DESIGNS****Date: 06.04.2018, Friday****Time: 10.00 am to 01.00 pm****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the question below. [11]**

- [A] What is Smart Appliances? Give one example of smart appliances. [2]
- [B] Why Real Time Operating System (RTOS) is used in embedded system? [3]
- [C] Explain various data-types of python programming. [3]
- [D] What is hardware timer and software timer? Give advantages and disadvantages of each type of timer. [3]

Q - 2 Answer any THREE. [12]

- [A] Draw only : Embedded system for digital camera.
- [B] Write an Arduino code to generate counter of 0 to 9 on seven-segment display with 1 sec time interval.
- [C] Explain the tradeoff between power and performance in embedded system.
- [D] What are the internal components of Embedded Hardware Units? Explain functionality of each component.

Q - 3 Answer any TWO. [12]

- [A] What are the difference between Embedded System & General purpose system?
- [B] Write an Arduino code to press any key from keypad & display it on LCD.
- [C] What are the design metric parameters of embedded system?

SECTION – II**Q - 4 Answer the question below. [11]**

- [A] State True/false: [2]
1. The turn-off delay is associated with a gate output transition to the high impedance value (z) from another value.
 2. The stimulus block is usually the top-level block.
- [B] Write down the difference between setup() & loop() function? [3]
- [C] Define: Rise delay, Fall delay and block comments [3]
- [D] Baud Data Rate is used in Serial Communication Why is it used? [3]

Q - 5 Answer any THREE. [12]

- [A] What is System on Chip? How does it differ from general-purpose system?
- [B] What is NRE cost and Unit cost?
- [C] What is Watchdog Timer? Where is it used?
- [D] Justify: Loss will increase if product is delayed in market.

Q - 6 Answer any TWO. [12]

- [A] Draw and explain the classification of Embedded System.
- [B] What are the major application areas of Embedded System
- [C] There are three key technologies for embedded systems namely Processor technology, IC technology and Design technology. How does IC technology impacts on embedded system?
